

Heston vs Black–Scholes: Fitting the Volatility Smile

One model prices every strike with the same number. The other doesn't have to.

The claim we test: Black-Scholes assumes a single, constant volatility — so it predicts the *same* implied vol at every strike. The market flatly disagrees: implied vol curves into a **smile/skew**. The Heston model, by letting volatility itself be random, can bend to fit that curve. This piece pits the two against a real SPX-style implied-vol surface and measures who fits, by how much, and at what cost.

PROJECT CARD

CATEGORY

Quant Foundations & Derivatives

LIBRARIES

QuantLib · SciPy · NumPy

ASSETS

SPX / SPY options

TIMEFRAME

2019 – 2024 + recent option chain

USE

Pricing options consistently across strikes — the vol desk's answer to the smile BS can't fit

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
plt.rcParams["figure.dpi"] = 110
```

1. The disagreement

Black-Scholes takes one volatility σ and returns one price per strike. Invert real option prices back into implied vols and — if BS were right — you'd get a **flat line**: the same σ at every strike. Instead, equity index options show a pronounced **skew**: deep out-of-the-money puts trade at much higher implied vol than at-the-money or upside calls.

Why? Two BS assumptions break:

- **Volatility isn't constant** — it clusters, spikes, and mean-reverts.
- **Returns aren't Gaussian** — crashes are fatter and more frequent than a normal distribution allows, and they cluster in falling markets.

The skew is the market pricing in exactly those two facts. Any single-vol model is structurally unable to reproduce it.

2. The contenders

	Black-Scholes	Heston
Volatility	Constant σ	Stochastic — its own mean-reverting process
Free parameters	1	5
Returns distribution	Log-normal (Gaussian log-returns)	Fat-tailed, skewed
Can fit the smile?	No — flat by construction	Yes
Closed form?	Yes (simple)	Semi-closed (characteristic function)

Black-Scholes assumes $dS = \mu S dt + \sigma S dW$ — the GBM from Tutorial 1.

Heston adds a second equation for variance v_t :

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S, \quad dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^\nu$$

with $\text{corr}(dW^S, dW^\nu) = \rho$. The five parameters: κ (mean-reversion speed), θ (long-run variance), ξ (vol-of-vol), ρ (spot-vol correlation, the skew driver), and v_0 (initial variance).

3. The test

We build a representative SPX implied-vol skew (ATM $\approx 20\%$, steep put wing — the shape index options actually trade) and give **both** models the same job: match it.

- **Black-Scholes** gets one degree of freedom — its best move is a single flat line, so we set it to the ATM vol.
- **Heston** gets calibrated: we search its five parameters to minimise the gap to the market smile.

Note: we use a representative parametric skew so the notebook runs without a live option-chain subscription; the calibration machinery is identical for a real SPX/SPY chain.

```
# --- representative SPX 3M implied-vol skew (moneyness K/S -> implied vol) ---
S0, T, r, q = 100.0, 0.25, 0.03, 0.0
moneyness = np.array([0.80, 0.85, 0.90, 0.95, 1.00, 1.05, 1.10, 1.15, 1.20])
strikes = moneyness * S0

# market smile: high vol in the put wing, gentle rise in far calls (typical equity skew)
mkt_iv = np.array([0.32, 0.285, 0.255, 0.228, 0.205, 0.192, 0.188, 0.191, 0.198])

fig, ax = plt.subplots(figsize=(9, 5))
ax.plot(moneyness, mkt_iv * 100, "o-", color="black", label="Market implied vol (SPX-style skew)")
ax.axhline(mkt_iv[4] * 100, color="steelblue", ls="--", label="Black-Scholes (single vol = ATM)")
ax.set_xlabel("Moneyness K / S"); ax.set_ylabel("Implied volatility (%)" )
```

```
ax.set_title("The disagreement: market skew vs the flat line BS must draw")
ax.legend(); plt.tight_layout(); plt.show()
```

Black-Scholes is a horizontal line. The market is a curve. That gap — biggest in the crash-protection put wing — is the mispricing a single-vol model bakes into every out-of-the-money option.

3.1 Calibrate Heston (QuantLib pricing + bounded optimisation)

A single-maturity smile under-determines five parameters — unconstrained optimisers happily wander into absurd values (κ in the thousands) that fit the curve for the wrong reasons. So we calibrate with **economically sensible bounds**: mean-reversion $\kappa \in [0.5, 8]$, vol-of-vol $\xi \leq 2$, correlation $\rho \in [-0.95, -0.05]$, variances ≤ 0.25 . QuantLib prices; SciPy searches.

```
import QuantLib as ql
from scipy.optimize import least_squares

today = ql.Date(15, 6, 2024)
ql.Settings.instance().evaluationDate = today
day_count = ql.Actual365Fixed()
calendar = ql.NullCalendar()

spot_h = ql.QuoteHandle(ql.SimpleQuote(50))
r_ts = ql.YieldTermStructureHandle(ql.FlatForward(today, r, day_count))
q_ts = ql.YieldTermStructureHandle(ql.FlatForward(today, q, day_count))
maturity = ql.Period(3, ql.Months)
expiry = today + maturity

def heston_ivs(params):
    """Price every strike under Heston, return Black-Scholes implied vols."""
    v0, kappa, theta, xi, rho = params
    proc = ql.HestonProcess(r_ts, q_ts, spot_h, v0, kappa, theta, xi, rho)
    eng = ql.AnalyticHestonEngine(ql.HestonModel(proc))
    bsm = ql.BlackScholesMertonProcess(spot_h, q_ts, r_ts,
                                       ql.BlackVolTermStructureHandle(
                                           ql.BlackConstantVol(today, calendar, 0.20, day_count)))
    ivs = []
    for K in strikes:
        opt = ql.VanillaOption(ql.PlainVanillaPayoff(ql.Option.Call, K),
                               ql.EuropeanExercise(expiry))
        opt.setPricingEngine(eng)
        try:
            ivs.append(opt.impliedVolatility(opt.NPV(), bsm))
        except Exception:
            ivs.append(np.nan)
    return np.array(ivs)

def residuals(p):
    return np.nan_to_num(heston_ivs(p) - mkt_iv, nan=0.5)

x0 = [0.042, 2.0, 0.045, 0.8, -0.7] # v0, kappa, theta, xi, rho
bounds = ([0.005, 0.5, 0.005, 0.05, -0.95], [0.25, 8.0, 0.25, 2.0, -0.05])
res = least_squares(residuals, x0, bounds=bounds, xtol=1e-12, ftol=1e-12)

v0, kappa, theta, xi, rho = res.x
print("Calibrated Heston parameters (bounded):")
print(f" v0 (initial var) = {v0:.4f} -> vol {np.sqrt(v0):.1%}")
print(f" theta (long-run var) = {theta:.4f} -> vol {np.sqrt(theta):.1%}")
print(f" kappa (mean-reversion) = {kappa:.2f}")
print(f" xi (vol-of-vol) = {xi:.2f}")
print(f" rho (spot-vol corr) = {rho:.2f} <- negative = downside skew")
```

3.2 Recover Heston's implied-vol smile

Price each strike under the calibrated model and invert to implied vol — same axis as the market curve.

```
heston_iv = heston_ivs(res.x)
bs_iv = np.full_like(mkt_iv, mkt_iv[4]) # flat at ATM

print("moneyness market heston BS(flat)")
for m, mk, he, bs in zip(moneyness, mkt_iv, heston_iv, bs_iv):
    print(f" {m:.2f}      {mk:6.1%} {he:6.1%} {bs:6.1%}")
```

4. The scoreboard

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(13, 5))

# left: the three curves
ax1.plot(moneyness, mkt_iv * 100, "o-", color="black", label="Market", lw=2)
ax1.plot(moneyness, heston_iv * 100, "s--", color="crimson", label="Heston (calibrated)")
ax1.plot(moneyness, bs_iv * 100, "-.-", color="steelblue", label="Black-Scholes (flat)")
ax1.set_xlabel("Moneyness K / S"); ax1.set_ylabel("Implied vol (%)")
ax1.set_title("Fit to the smile"); ax1.legend()

# right: absolute pricing error in vol points
bs_err = np.abs(bs_iv - mkt_iv) * 100
he_err = np.abs(heston_iv - mkt_iv) * 100
x = np.arange(len(moneyness)); w = 0.38
ax2.bar(x - w/2, bs_err, w, color="steelblue", label="Black-Scholes")
ax2.bar(x + w/2, he_err, w, color="crimson", label="Heston")
ax2.set_xticks(x); ax2.set_xticklabels([f"{m:.2f}" for m in moneyness], rotation=45)
ax2.set_xlabel("Moneyness K / S"); ax2.set_ylabel("|error| (vol points)")
ax2.set_title("Absolute implied-vol error by strike"); ax2.legend()
plt.tight_layout(); plt.show()

print(f"Mean abs error – Black-Scholes: {bs_err.mean():.2f} vol pts")
print(f"Mean abs error – Heston: {he_err.mean():.2f} vol pts")
print(f"Worst strike (put wing) – BS: {bs_err[0]:.2f} pts vs Heston: {he_err[0]:.2f} pts")
```

The picture is decisive where it matters most — the **put wing**. Black-Scholes underprices crash protection by a wide margin (its flat line sits far below the market's elevated put vols), while Heston tracks the curve because its negative ρ generates exactly that downside skew. Across the surface Heston's mean error is a fraction of Black-Scholes'.

5. Verdict

Heston wins the fit — decisively — and it isn't close in the wings. With five parameters versus one, that's expected; the real question is whether the extra complexity earns its keep. The answer depends on the job:

- **Pricing a single vanilla at one strike?** Black-Scholes with that strike's *own* implied vol is simpler and exact. The smile is a lookup table; you don't need a model to read one row.
- **Pricing a book across many strikes consistently, or anything exotic (barriers, cliquets, forward-starts) whose value depends on the *dynamics* of vol?** Heston is close to essential — a flat vol would misprice the smile-sensitive payoff.

What each model really is:

- Black-Scholes isn't wrong so much as *incomplete* — it's the quoting convention (implied vol) more than a dynamic model. Its flat smile is a feature of using one number, not a bug to be fixed.
- Heston buys smile-consistency and fat tails at the cost of five parameters that must be **recalibrated** as the surface moves, and a fit that can still miss very short maturities (where jumps, not stochastic vol, drive the steep skew).

The honest trade-off: more parameters always fit better in-sample. Heston earns trust because its parameters are *economically interpretable* (ρ is the skew, ξ the smile convexity, θ the vol term structure) — it's fitting the smile *for the right reasons*, not just curve-fitting.

Where to go next: for the very short-dated skew Heston struggles with, add jumps (**Bates**, Heston + Merton jumps); to fit the *entire* observed surface exactly rather than approximately, **Dupire local volatility**; and for the constant-vol baseline both are measured against, revisit the **Black-Scholes** tutorial.